1️⃣ What is NSG?

A **Network Security Group (NSG)** in Azure is a Layer-4 (TCP/UDP) firewall that controls inbound and outbound traffic for:

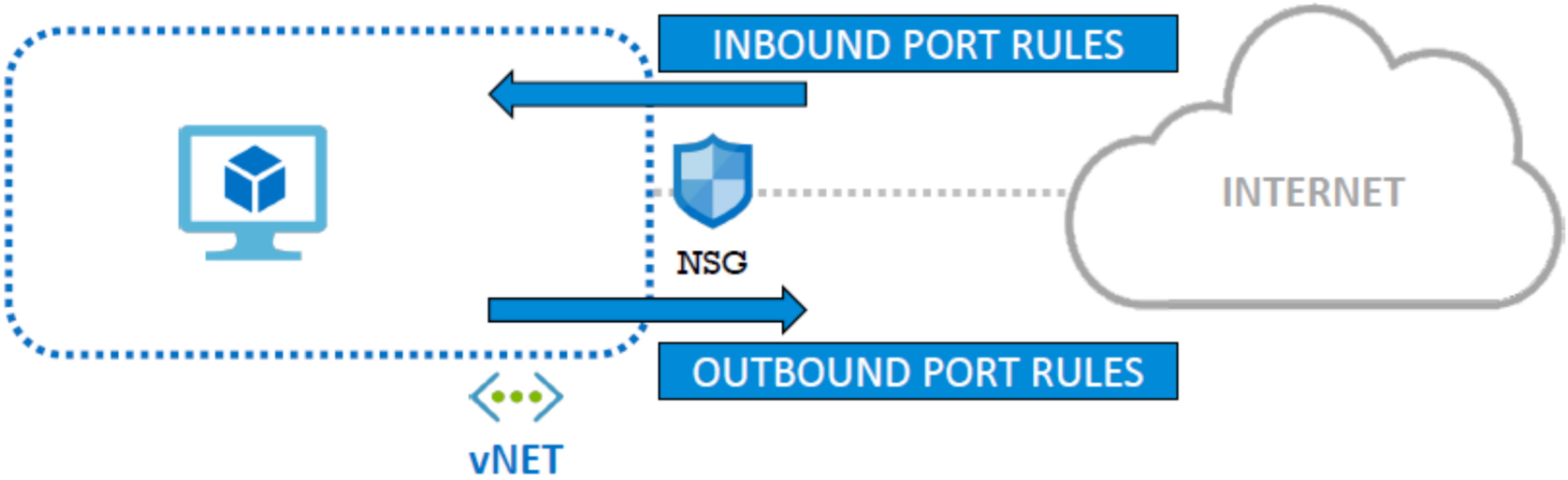
- Subnets
- Network Interfaces (NICs)

It filters traffic based on:

- Source IP
- Destination IP
- Port
- Protocol (TCP/UDP/Any)
- Direction (Inbound/Outbound)

Network security groups

Network Security Group (NSG)



2️⃣ Where Can NSG Be Applied?

Level	Scope	Use Case
Subnet	Applies to all VMs in subnet	Common security rule
NIC	Applies to specific VM only	Fine-grained control

💡 If both subnet + NIC NSG exist → **Both rules are evaluated**

3️⃣ How NSG Works (Rule Processing Logic)

Rule Evaluation Order

1. Lowest **Priority number** is evaluated first.
2. Once matched → Processing stops.
3. Default rules always exist.

Priority Range:

- 100 – 4096
- Lower number = Higher priority

4️⃣ Default NSG Rules (Very Important for AZ-104)

Inbound Default Rules

Priority	Name	Action
65000	AllowVnetInBound	Allow
65001	AllowAzureLoadBalancerInBound	Allow
65500	DenyAllInBound	Deny

Outbound Default Rules

Priority	Name	Action
65000	AllowVnetOutBound	Allow
65001	AllowInternetOutBound	Allow
65500	DenyAllOutBound	Deny

⚠️ You cannot delete default rules.

You override them using lower priority custom rules.

5️⃣ NSG Rule Components (Exam Important)

Field	Meaning
Source	IP / Service Tag / ASG
Source Port	Usually *
Destination	IP / Service Tag / ASG
Destination Port	Example: 80, 443, 3389
Protocol	TCP / UDP / Any
Action	Allow / Deny
Priority	100–4096

6️⃣ Service Tags (Very Important)

Examples:

- Internet
- VirtualNetwork
- AzureLoadBalancer
- Storage
- SQL

Used to simplify rule creation.

7️⃣ Application Security Groups (ASG)

Used to group VMs logically.

Example:

- WebServers ASG
- AppServers ASG
- DBServers ASG

Instead of IP-based rules → use ASG.

🔬 LAB 1 – Basic NSG Inbound Rule (Allow RDP Only)

🎯 Scenario

Allow RDP (3389) from your public IP only. Block others.

Step 1: Create VM

Portal → Virtual Machines → Create

Allow RDP during creation (temporary)

Step 2: Create NSG

Portal → Network Security Groups → Create

- Name: NSG-RDP-Lab
- Region: Same as VM

Step 3: Add Inbound Rule

Inbound Security Rules → Add

- Source: IP Addresses
- Source IP: Your Public IP
- Destination: Any
- Port: 3389
- Protocol: TCP
- Action: Allow
- Priority: 100

Click Save.

Step 4: Associate NSG

VM → Networking → Attach NSG to NIC or Subnet.

Step 5: Test

✔️ RDP from your IP → Should Work

❌ RDP from other IP → Blocked

🔬 LAB 2 – Deny Internet Access (Outbound Rule)

🎯 Scenario

Block VM from accessing Internet.

Add Outbound Rule

Outbound → Add Rule:

- Source: Any
- Destination: Service Tag → Internet
- Port: *
- Protocol: Any
- Action: Deny
- Priority: 100

Save.

Test

Inside VM:

```nginx
ping google.com
```

❌ Should fail.

🔬 LAB 3 – Subnet to Subnet Communication Control

🎯 Scenario

Subnet A cannot talk to Subnet B.

1. Create VNet with:
    - Subnet-Frontend
    - Subnet-Backend
2. Deploy 1 VM in each subnet.
3. Attach NSG to Backend Subnet.

4. Add Rule:
    - Source: Subnet-Frontend CIDR
    - Destination: Any
    - Port: *
    - Action: Deny
    - Priority: 100

Test using:

```cpp
ping private-ip
```

Should fail.